

RECSYS CHALLENGE 2026 · MUSIC-CRS

# State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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47,071

tracks in the catalog

11

ways to search

1

learned ranker





# The short version

SUMMARY

TALK ONLY

## WHAT WE BUILT

An LLM reads each turn into a **typed state**; eleven retrieval branches search one index; a **LightGBM** ranker picks 20; an LLM writes the reply.

## HOW IT SCORED

# 0.3811

Blind-B · 29th of 40  
nDCG@20 fell from 0.438 on Blind-A to 0.254.

## WHAT WE FOUND

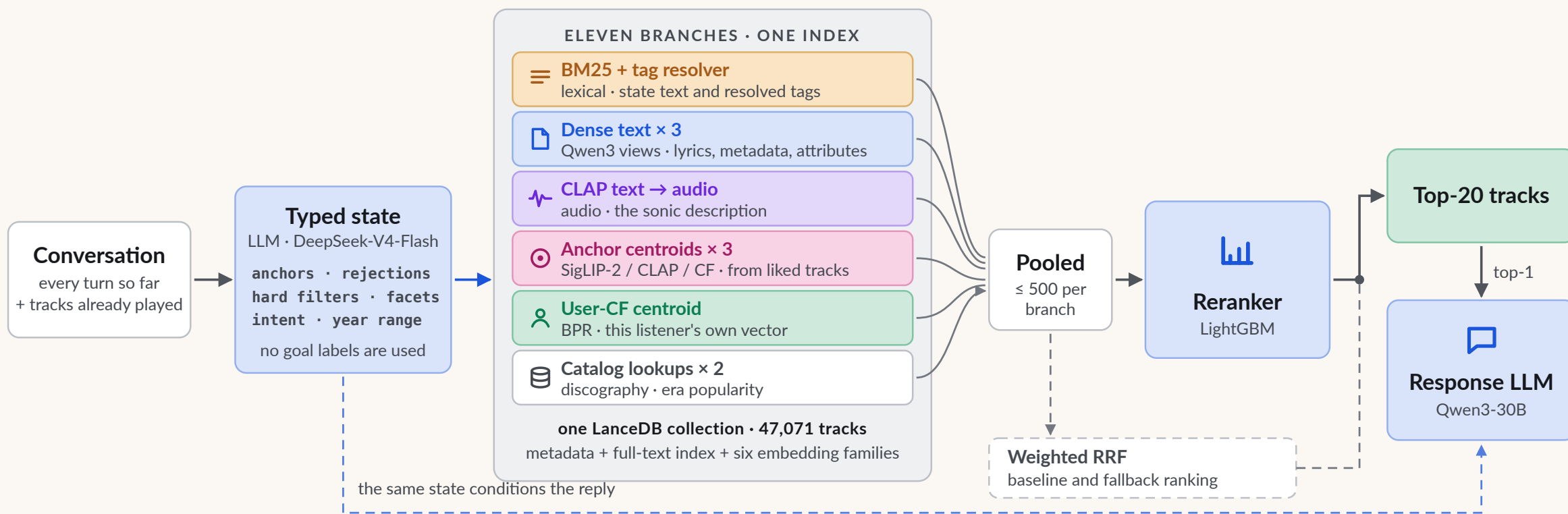
# 31 of 43

diagnosed failures were ranking

Extraction was **3**. The state was richer than what the ranker used.

The system understood the conversation more often than it **acted on it**.

# One state, eleven branches, one learned ranker



Organisers supplied the audio (CLAP), cover-art (SigLIP-2), listening-data (BPR) and small text (Qwen3-0.6B) embeddings; we added Qwen3-8B text embeddings and our own bi-encoder.



# Request understanding

TURN 1



“Calling Cards” — *not quite*

TURN 2



“Man” — *not it either*

TURN 3 · NOW

Much more subdued... a sense of place in the lyrics... almost a cappella.

LLM



STAY WITH

Neko Case

TURNED DOWN

“Calling Cards”, “Man”

MOOD

subdued

LYRICS

a sense of place

SOUND

stark, almost a cappella

The LLM reads **every turn so far** and writes structured fields; every later step reads those.



# Retrieval — eleven branches

APPROACH

CORE

×1



## BM25 + tag resolver

keywords and resolved tags  
from the state

×3



## Dense text

Qwen3 embeddings of lyrics,  
metadata, attributes

×1



## CLAP text → audio

the sonic description, matched  
to audio

×3



## Anchor centroids

near liked tracks: cover art,  
audio, listening

×1



## User-CF centroid

this listener's own BPR vector

×2



## Catalog lookups

the artist's discography; era  
popularity

Each branch returns its **top 500**; the ranker sees their union.



# What the ranker leans on

| FEATURE FAMILY              | #  | GAIN  |
|-----------------------------|----|-------|
| Bi-encoder cosine           | 1  | 59.0% |
| Per-branch rank and score   | 66 | 11.5% |
| Session and artist affinity | 18 | 10.7% |
| Popularity · state · tags   | 31 | 9.2%  |
| Other similarity cosines    | 17 | 8.9%  |
| Constraints and rejections  | 13 | 0.7%  |

## SETUP

LightGBM LambdaMART, a gradient-boosted tree ranker. LR 0.025, 127 leaves, truncation 200, 5-fold user-grouped CV. ~20.6M candidate rows.

## READ IT CAREFULLY

Split gain describes the fitted model, not recommendation quality. But **constraints at 0.7%** is real: the ranker leans on similarity, not on what the listener asked for.

## GUARDS AFTER SCORING

Exact-track pin when a specific track is named; a final artist-constraint check.



# How it scored

| OFFICIAL CODABENCH SPLIT    | NDCG@20 | CAT.   | LEX.   | JUDGE | COMPOSITE |
|-----------------------------|---------|--------|--------|-------|-----------|
| Blind-A · development phase | 0.4380  | 0.0313 | 0.8028 | 4.70  | 0.5799    |
| Blind-B · final             | 0.2537  | 0.0315 | 0.7862 | 3.30  | 0.3811    |

NDCG@20

-42%

0.4380 → 0.2537

JUDGE (1-5)

-1.40

4.70 → 3.30

DIVERSITY TERMS

flat

catalog and reply-wording  
diversity barely moved

Blind-B also withholds the conversation-goal annotations — but the deployed ranker never used them at inference, so that is not the explanation.



# Extracted, then ignored

WENT WRONG

CORE



*"I gotta break away from **Deltron** and **Del** completely"*

**Dead Prez — Hip Hop**

state held both artists as avoid

IGNORED



*"branching out to **other** singer-songwriter artists"*

**Ryan Adams — When The Stars Go Blue**

state held Ryan Adams as avoid

IGNORED



*"still focusing on **Juanes**... haven't found the depth I'm after"*

**Juanes — Me Enamorado De Ti**

state held Juanes as rejected

IGNORED

Three sessions, one failure: the listener asked to move on, the state recorded it, and **the ranker served that artist anyway.**



# Where the failures were — all 80 Blind-B turns

ANALYSIS CORE

| DIAGNOSED GAP          | TURNS | MEANING                                                |
|------------------------|-------|--------------------------------------------------------|
| Ranking                | 31    | a better, non-violating candidate sat below our top-20 |
| Constraint not applied | 8     | an extracted filter never reached the pool             |
| State extraction       | 3     | the extractor missed something the user said           |
| Entity resolution      | 1     | a name never resolved to a catalog ID                  |

TOP-1 FLAGGED

17 / 80

the track we led with violated something the state had recorded

HOW THIS WAS COUNTED

43 diagnoses across 32 distinct turns. A **rule-based check** of each turn’s system trace — not an LLM, and not a relevance label.

Extraction is **3 of 43** diagnoses. The failure is overwhelmingly downstream of understanding the conversation.



# Asked for, served

WENT WRONG

TALK ONLY

1

*“the Kamelot track from **Silverthorn**”*

**Kamelot — Fallen Star**

from *Haven*. We extracted “Silverthorn” but never filtered on it.

CONSTRAINT

2

*“exactly **126.70 bpm**, in **G minor**”*

**no tempo or key in the catalog**

and we added no data from outside it

NO DATA

3

*“**‘Watercolors’** by **Pat Metheny**”*

**Pat Metheny — Alfie**

reply: “Got it — here’s ‘Alfie’...” — “Watercolors” is not in the catalog

RESPONSE



# What the training labels look like

WENT WRONG

CORE



*“I keep asking for Arcade Fire — that's the only song I want”*

label: Arctic Monkeys — This House is a Circus

DOES NOT



*“similar chill vibe, but from a different artist”*

label: Bonobo — Jets

MOVES



*“what **other artists** fit this darkwave style?”*

label: Tropic of Cancer — The Dull Age

MOVES



*“what **other** orchestral tracks would you recommend?”*

label: Nick Phoenix — Kogan

MOVES

Three of four are annotated **MOVES TOWARD GOAL** — the dataset records them as having helped. Train on this and the model learns to anchor. **Ours does.**

Why: the answer is the simulated recommender's pick from a 16–32-track pool, made without seeing the listener's goal.



# Everything is released

THANK YOU

CORE



Paper, poster, slides, code, models, data

[npatta01.github.io/music-crs-2026](https://npatta01.github.io/music-crs-2026)